CS109A — Lecture 11: Intro to Classification and Logistic Regression

CS109A Introduction to Data Science

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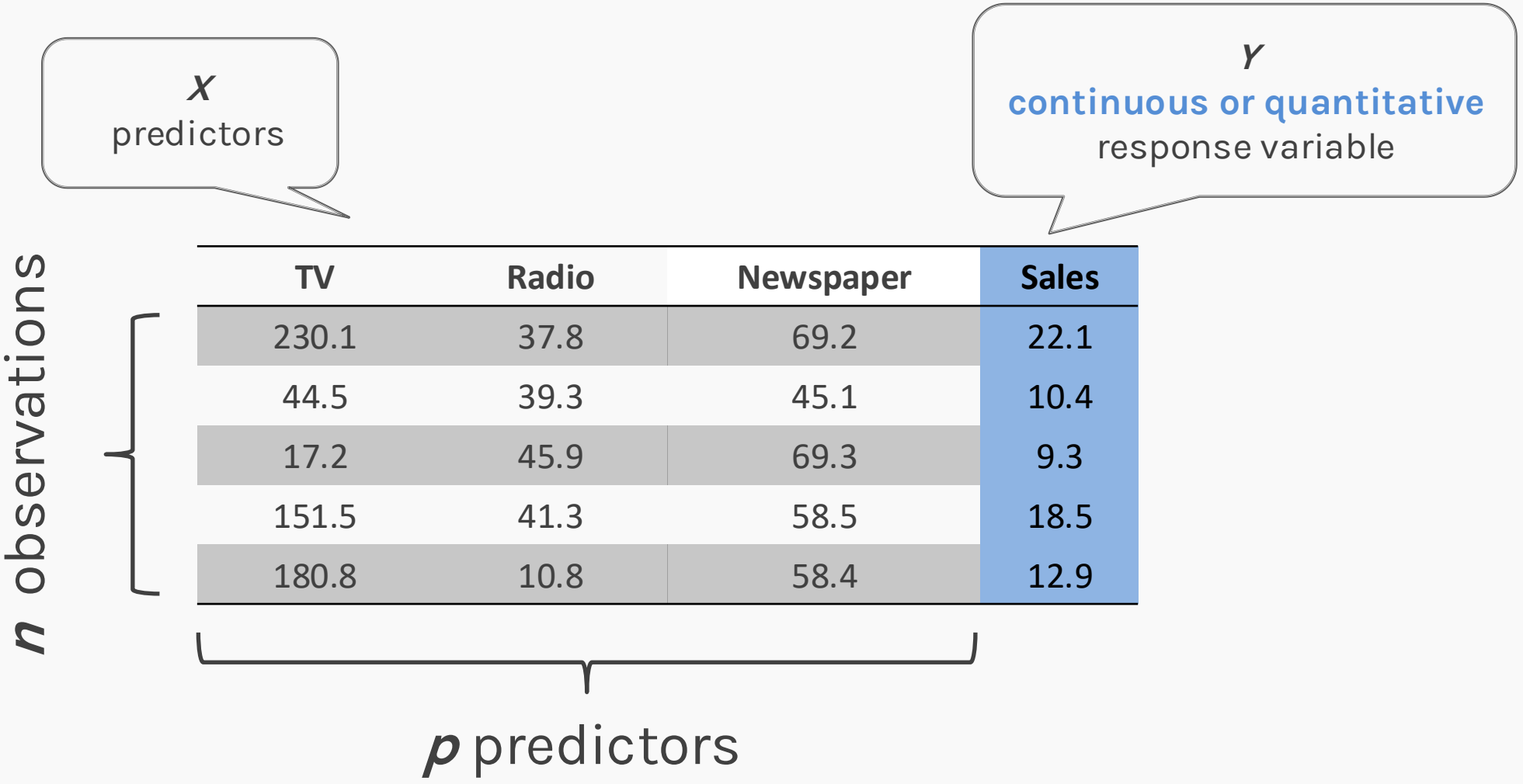
Old Orchard Beach, ME

Outline

- **What is Classification?**
- Why not Linear Regression?
- Estimating the Simple Logistic Model
- Multiple Logistic Regression
- Classification Decision Boundaries

What is Classification?

Advertising Data (from earlier lectures)



TV	Radio	Newspaper	Sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	9.3
151.5	41.3	58.5	18.5
180.8	10.8	58.4	12.9

What is Classification?

Consider another dataset that contains a binary outcome for acute heart disease **(AHD)** for 303 patients who presented with chest pain.

X
predictors

Y
Yes or No
response variable

Age	Sex	ChestPain	RestBP	Chol	MaxHR	ExAng	Thal	AHD
63	1	typical	145	233	150	0	fixed	No
67	1	asymptomatic	160	286	108	1	normal	Yes
67	1	asymptomatic	120	229	129	1	reversable	Yes
37	1	nonanginal	130	250	187	0	normal	No

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Yes indicates **presence** of heart disease

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No indicates **absence** of heart disease

What is Classification?

In summary,

Regression

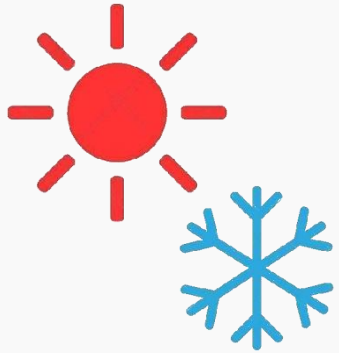
Performs well on tasks that require the prediction of a quantitative response variable.



What is the temperature going to be tomorrow?

Classification

Performs well on tasks that require the prediction of a categorical or qualitative response variable. It classifies an observation into a category or class labeled by Y.



Is it going to be hot or cold tomorrow?

Typical Classification Examples

Classification problems are ubiquitous in many domains, such as healthcare, finance, and sports.

Some examples of classification problems are:

- To determine whether a startup is worth investing in
- To determine the disease type of patients based on various genomic markers
- To determine if a user is more likely to click on an advertisement.
- To determine if a given image is a real or a fake one
- To determine which team will win a game in any 1-v-1 sport.

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- What is Classification?
- **Why not Linear Regression?**
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Why not Linear Regression?

Assume you are given a dataset containing information of different students and your task is to predict whether their major is Computer Science, Statistics or otherwise.

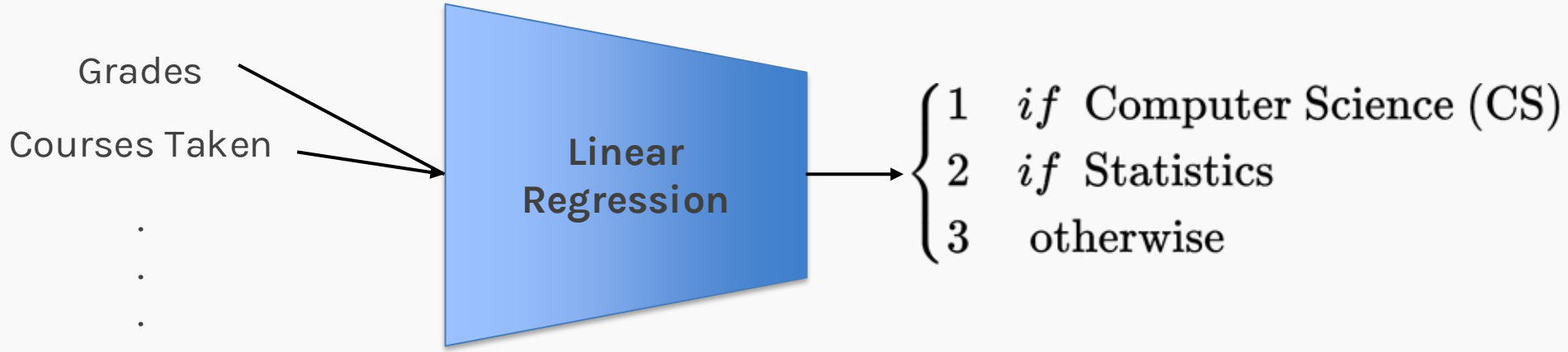
This categorical variable can't be used as is, but it could be encoded to be quantitative.

If y represents majors, then it could take on the values:

$$y = \begin{cases} 1 & if \text{ Computer Science (CS)} \\ 2 & if \text{ Statistics} \\ 3 & \text{otherwise} \end{cases}$$

Why not Linear Regression?

Now that we have encoded the values, a linear regression could be used to predict y from x.



But what is the problem here?

Why not Linear Regression?

This model would imply a specific ordering of the outcome.

For example, a change from $y = 1$ to $y = 2$ (Computer Science to Statistics) is the considered the same as a change from $y = 2$ to $y = 3$ (Statistics to everyone else). However, this change should not be interpreted as the same.

If a categorical response variable is **ordinal** (has a natural ordering, like Freshman, Sophomore, etc.), then a linear regression model would make some sense but is still not ideal.

Why not Linear Regression?

Additionally, if the ordering of the response variable is changed, the model estimates and predictions would be fundamentally different.

For example, a model trained with $y = 1$ represents Statistics and $y = 2$ represents CS is different from a model trained with the original ordering.

Why not Linear Regression?

- Consider a simpler problem where the response variable y has only two categories. Here, there is a natural ordering of the categories.

$$y = \begin{cases} 1, & has\ heart\ disease \\ 0, & no\ heart\ disease \end{cases}$$

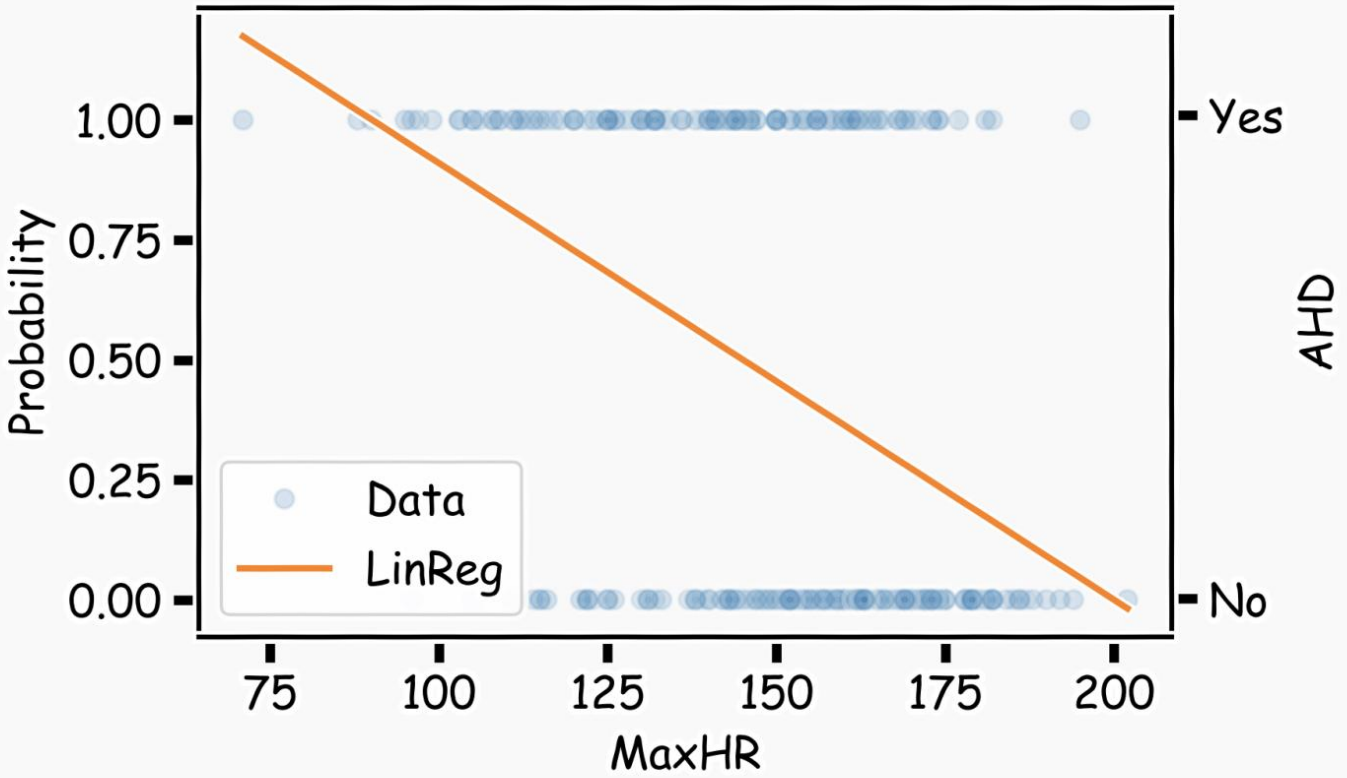
- Linear regression could be used to predict the probability $P(y = 1)$ directly from a set of predictors such as sex, cholesterol levels, etc.

- If $P(y = 1) \geq 0.5$, we could predict that the patient has heart disease and predict otherwise if $P(y = 1) < 0.5$.

Why not Linear Regression?

What could go wrong with this linear regression model?

Since this is modeling $P(y = 1)$, values for $\hat{y}$ below 0 and above 1 would not make sense as a probability.

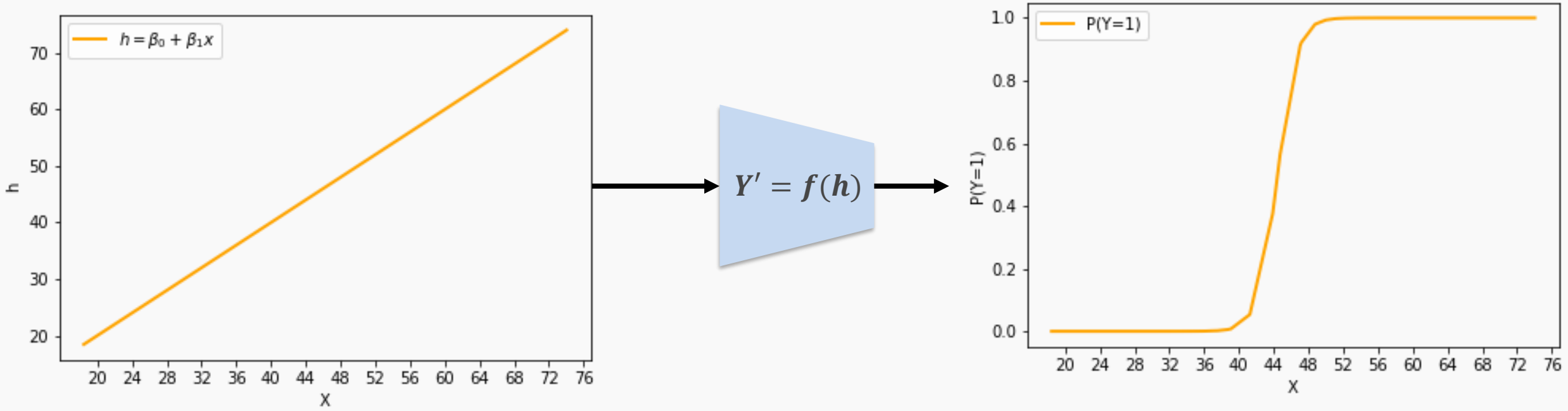


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What function should we use?

Now we know that linear regression yields values for probability that are larger than 1 or smaller than 0. So what can we do to fix this?



What function should we use?

We can use the sigmoid function:

$$h = \beta_0 + \beta_1 X \longrightarrow \mathrm{p} = \frac{1}{1 + e^{-h}} \longrightarrow P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

Logistic Regression



- Logistic Regression addresses the problem of estimating a probability, $P(y = 1)$, to be outside the range of $[0,1]$.

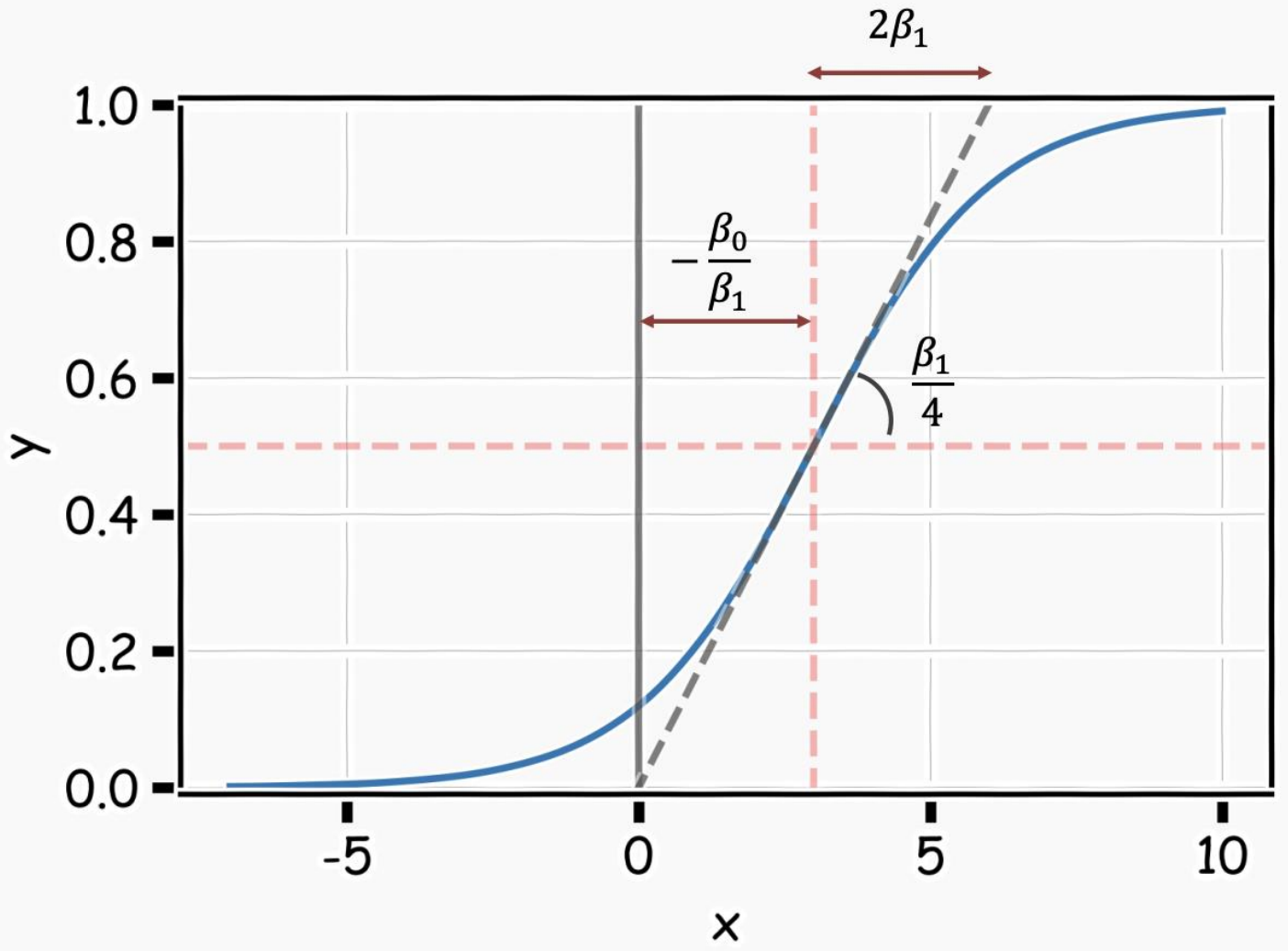
- The logistic regression model uses a function, called the **logistic** function, to model $P(y = 1)$:

$$P(Y = 1) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}} = \frac{1}{1 + e^{-(\beta_0+\beta_1 X)}}$$

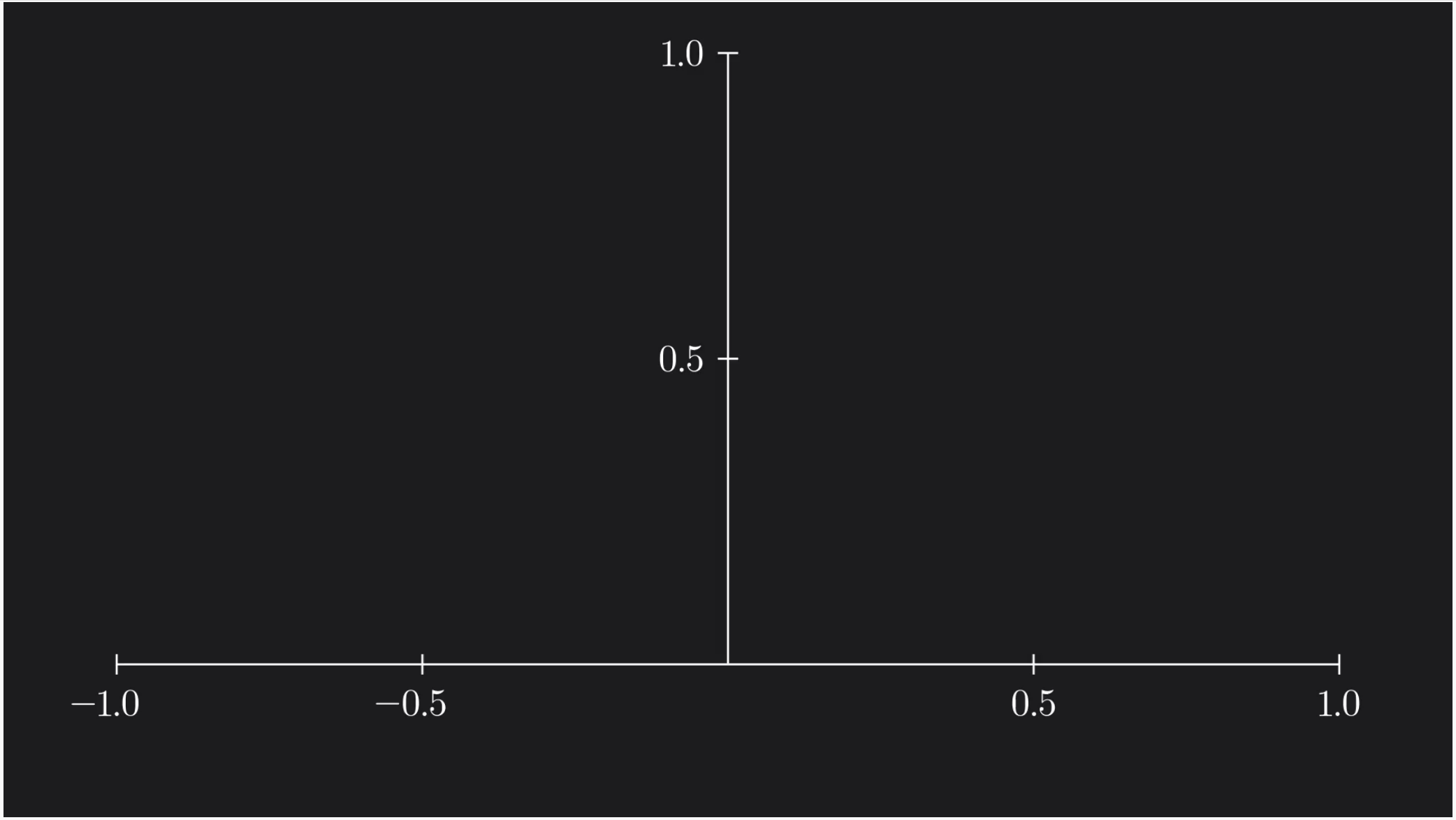
What are the parameters of this model?

Logistic Regression

The coefficients β_0 and β_1 now control the shape of this *S*-shaped curve.



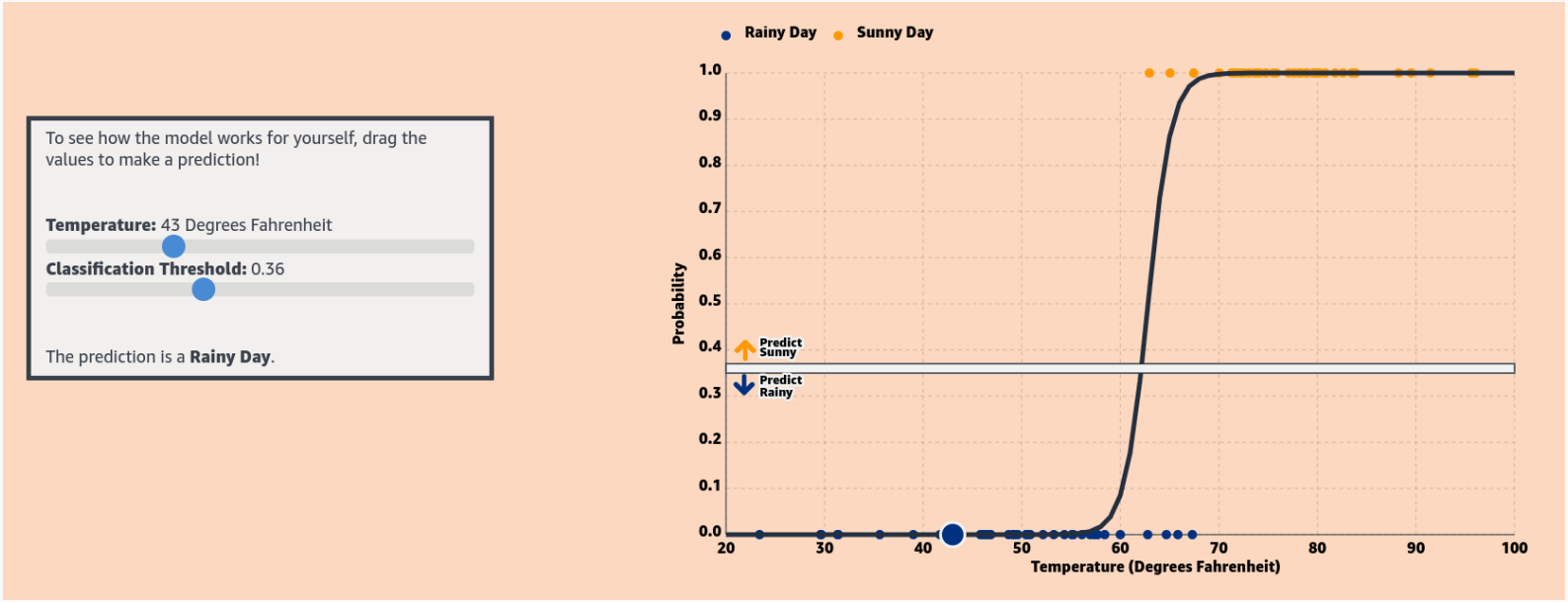
Sigmoid Animation



Side Note: MLU-Explain

MLU-Explain from Amazon hosts many interactive visual explanations of core machine learning concepts. Their page for logistic regression is a great resource!

https://mlu-explain.github.io/logistic-regression/



Interpretation of β's

With a little bit of algebraic work, the logistic model can be rewritten as:

$$\ln\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1 X$$

odds

Logistic regression is said to model the ***log-odds*** with a linear function of the predictors or features, X.

A one-unit change in X is associated with a β_1 change in the log-odds of $\mathrm{P}(Y=1)$; or better yet, a one-unit change in X is associated with an e^{β_1} change in the odds that $Y=1$.

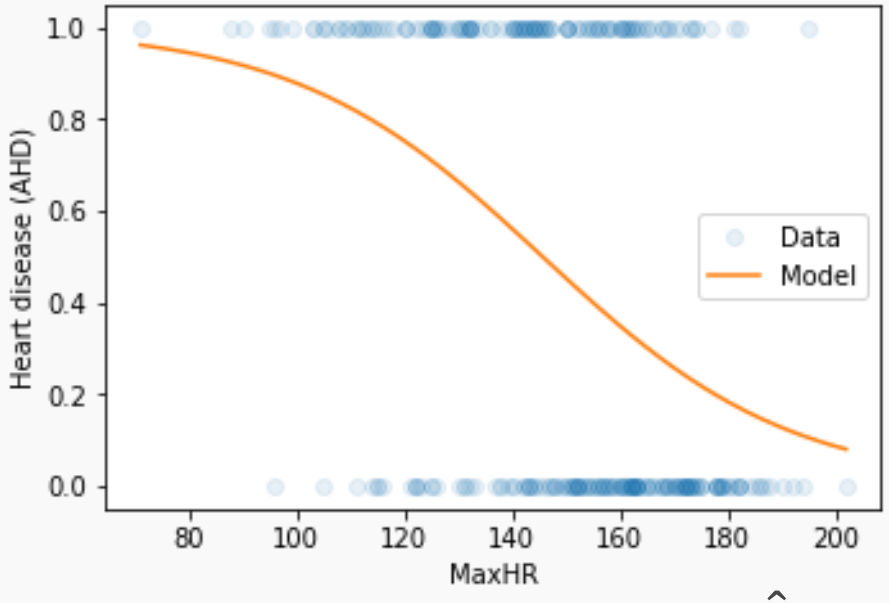
A First Logistic Regression Model in sklearn

Here is a logistic regression output to predict Y = AHD from X = MaxHR:

```
logreg = LogisticRegression(penalty='none')
logreg.fit(df_heart[['MaxHR']], df_heart['AHD'])

print('Estimated beta1: \n', logreg.coef_)
print('Estimated beta0: \n', logreg.intercept_)
```

```
Estimated beta1:
 [[-0.04341112]]
Estimated beta0:
 [6.3249492]
```



What is the estimated model? What are the interpretations of the $\hat{\beta}$s?

$$\ln\left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)}\right) = 6.325 - 0.0434(MaxHR)$$

Estimating the Simple Logistic Model

Estimating parameter coefficients

Unlike in linear regression where there exists a closed-form solution to finding the estimates, $\hat{\beta}_j$'s, that minimize the loss function, logistic regression estimates cannot be calculated through simple matrix multiplication.

Questions:

- In linear regression what loss function was used to determine the parameter estimates?

  MSE

- Should we use MSE in logistic regression? Why?

- What could/should we use instead?

Binary Cross-Entropy

The loss function used in logistic regression is called **Binary Cross Entropy**:

$$L_{BCE} = -\sum_i y_i \ln p_i + (1 - y_i) \ln(1 - p_i)$$

Where did β_0, β_1 go? What is p_i?

$$p_i = \frac{e^{\beta_0 + \beta_1 X_i}}{1 + e^{\beta_0 + \beta_1 X_i}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_i)}}$$

Binary Cross-Entropy: how?

The loss function used in logistic regression is called **Binary Cross Entropy**:

$$L_{BCE} = -\sum_i y_i \ln p_i + (1 - y_i) \ln(1 - p_i)$$

$$\text{where } p_i = \frac{e^{\beta_0 + \beta_1 X_i}}{1 + e^{\beta_0 + \beta_1 X_i}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_i)}}$$

How do we go about minimizing this loss function?

We take the derivatives (with respect to β_0 and β_1) and set to zero!

Yuck, that's no fun! We gotta use the chain rule…and we are not guaranteed of a closed form solution ☹

So where do we turn? An iterative algorithm like Gradient Decent :)

Binary Cross-Entropy: why?

The loss function used in logistic regression is called **Binary Cross Entropy**:

$$L_{BCE} = -\sum_i y_i \ln p_i + (1 - y_i) \ln(1 - p_i)$$

$$\text{where } p_i = \frac{e^{\beta_0 + \beta_1 X_i}}{1 + e^{\beta_0 + \beta_1 X_i}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_i)}}$$

Where does this come from?

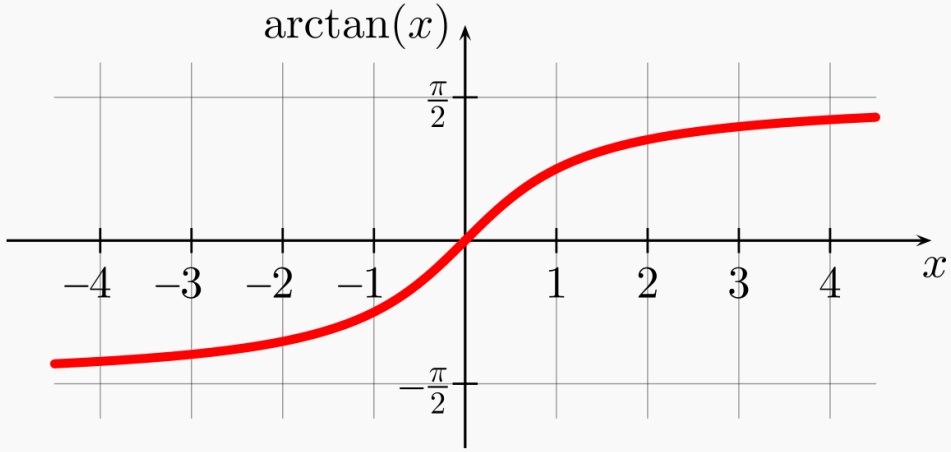
For my Stats friends: what log-PDF does this remind you of?
*Hint: what distribution should a binary y_i random variable follow?

Let $y_i \sim \text{Bernoulli}\left(p_i = \frac{e^{\beta_0 + \beta_1 X_i}}{1 + e^{\beta_0 + \beta_1 X_i}}\right)$. More on this later ☺

Food for thought:

Who came up with logistic regression? Why the logistic function again?

Why is the logistic function used as the S-shaped curve? What other functions could be used?



Stat 110 Idea: **any** CDF function (unbounded and continuous) could be used as the S-shaped curve. Econometricians love to use $\Phi(\beta_0 + \beta_1 X_1 + \cdots)$.

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Multiple Logistic Regression

It is simple to illustrate examples in logistic regression when there is just one predictors variable.

But the approach ‘easily’ generalizes to the situation where there are multiple predictors.

A lot of the same details as linear regression apply to logistic regression. Interactions can be considered, multicollinearity is a concern and so is overfitting.

So how do we correct for such problems?

Regularization and checking though train and cross-validation!

We will get into the details of this, along with other extensions of logistic regression, in the next lecture.

Multiple Logistic Regression

Earlier we saw the general form of *simple* logistic regression, meaning when there is just one predictor used in the model:

$$\ln\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1 X$$

Multiple logistic regression is a generalization to multiple predictors. More specifically we can define a multiple logistic regression model to predict $P(Y=1)$ as such:

$$\ln\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + .. + \beta_p X_p$$

Fitting Multiple Logistic Regression

The estimation procedure is identical to that as before for simple logistic regression:

- All parameters $\beta_0, \beta_1, \ldots, \beta_p$ are estimated using BCE as the loss function and using an iterative method like Gradient Descent.

The actual fitting of a Multiple Logistic Regression is easy using software (of course there's a python package for that) as the iterative minimization of BCEhas already been hard coded.

In the `sklearn.linear_model` package, you just have to create your multidimensional design matrix X to be used as predictors in the `LogisticRegression` function.

Interpretation of Multiple Logistic Regression

Interpreting the coefficients in a multiple logistic regression is similar to that of linear regression.

Key: since there are other predictors in the model, the coefficient $e^{\widehat{\beta}_j}$ is the multiplicative change in odds between the jth predictor and the response. But do we have to say "Controlling for the other predictors in the model"?

We are trying to attribute the partial *effects* of each model controlling for the others (aka, controlling for possible *confounders*).

Multicollinearity plays a role just like in linear regression.

2D Classification in Logistic Regression: Example #1

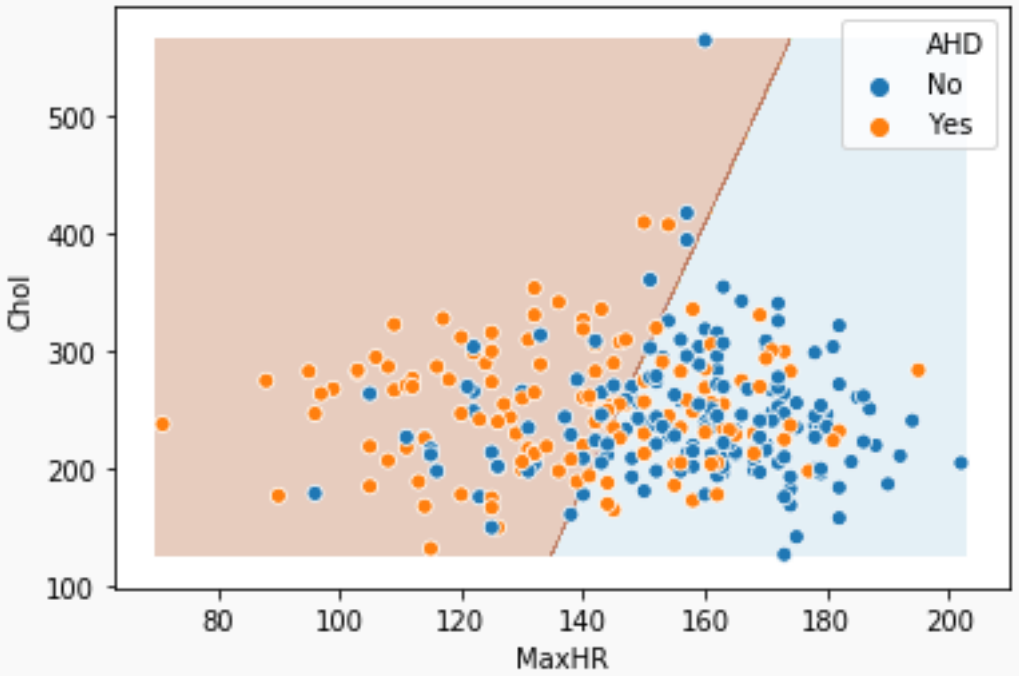
A logistic regression model was fit to predict Y = AHD from X_1 = MaxHR and X_2 = Chol. Results shown below:

```
data_x = df_heart[['MaxHR','Chol']]
data_y = df_heart['AHD']

logreg = LogisticRegression(penalty='none', fit_intercept=True)
logreg.fit(data_x, data_y);


print('Estimated beta1, beta2: \n', logreg.coef_)
print('Estimated beta0: \n', logreg.intercept_)
```

```
Estimated beta1, beta2:
 [[-0.04388093  0.00391746]]
Estimated beta0:
 [5.42271131]
```



What is the estimated model?

$$\ln\left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)}\right) = 5.423 - 0.0439(MaxHR) + 0.0039(Chol)$$

What are the interpretations of the $\hat{\beta}$s?

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Using Logistic Regression for Classification

How can we use logistic regression to perform classification?

That is, how can we predict when $Y = 1$ vs. when $Y = 0$?

We can classify all observations for which:

- Classify all observations with $\hat{P}(Y = 1) \geq 0.5$ to be in the group associated with $Y = 1$.
- Classify all observations with $\hat{P}(Y = 1) < 0.5$ to be in the group associated with $Y = 0$.

How would this extend if Y has 3+ classes?

Decision Boundaries for Classification

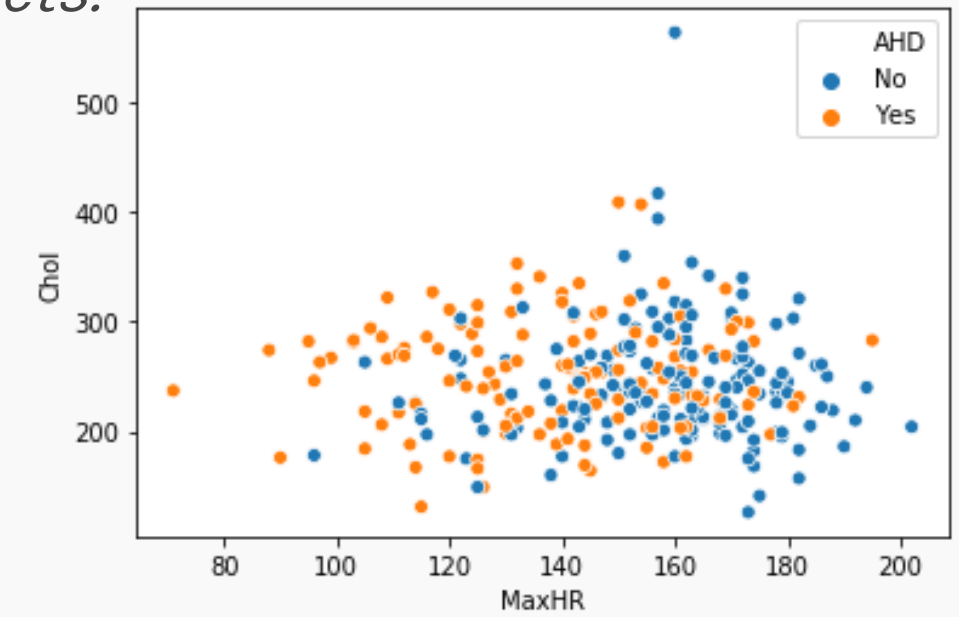
Recall that we could attempt to purely classify each observation based on whether the estimated $P(Y = 1)$ from the model is at least 0.5:

$$\hat{P}(Y = 1) \geq 0.5$$

This results in a **decision boundary**: a surface (line, curve, etc. in 2D) that separates the predicted classes into *sets.*

Here's a 2-D plot from our Heart Data Set:

How do you expect logistic regression to draw the decision boundary?



Decision Boundaries Example

Here is the output from the logistic regression model with 2 predictors:

What is the estimated model?

```
data_x = df_heart[['MaxHR','Chol']]
data_y = df_heart['AHD']

logreg = LogisticRegression(penalty='none', fit_intercept=True)
logreg.fit(data_x, data_y);


print('Estimated beta1, beta2: \n', logreg.coef_)
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Estimated beta1, beta2:
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Estimated beta0:
 [5.42271131]
```

$$\ln\left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)}\right) = 5.423 - 0.0439(MaxHR) + 0.0039(Chol)$$

What will the decision boundary look like? Key: if $\hat{P}(Y=1) = 0.5$, then what are the estimated odds? What are the estimated log-odds?

In logistic regression, the decision boundary is defined when $X\beta = 0$.

2D Classification in Logistic Regression: Example #1

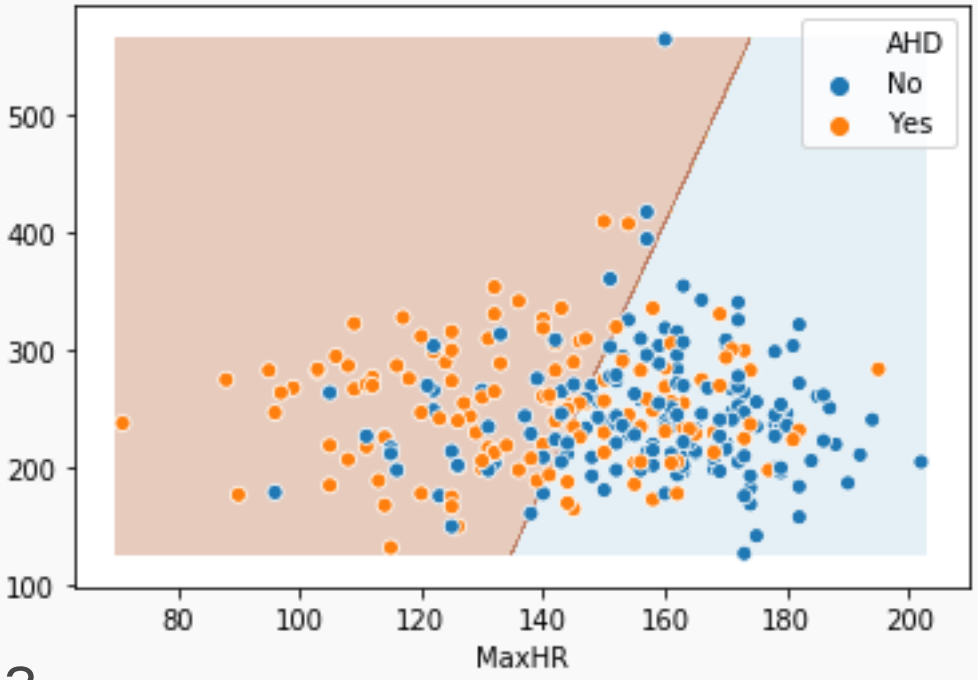
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Estimated beta0:
 [5.42271131]
```



What will the decision boundary look like?

In logistic regression, decision boundaries are structured to be linear!

2D Classification in Logistic Regression: Example #2

A logistic regression model was fit to predict Y = AHD from X_1 = MaxHR and X_2 = Chol, and X_3 = their interaction. Results are shown below:

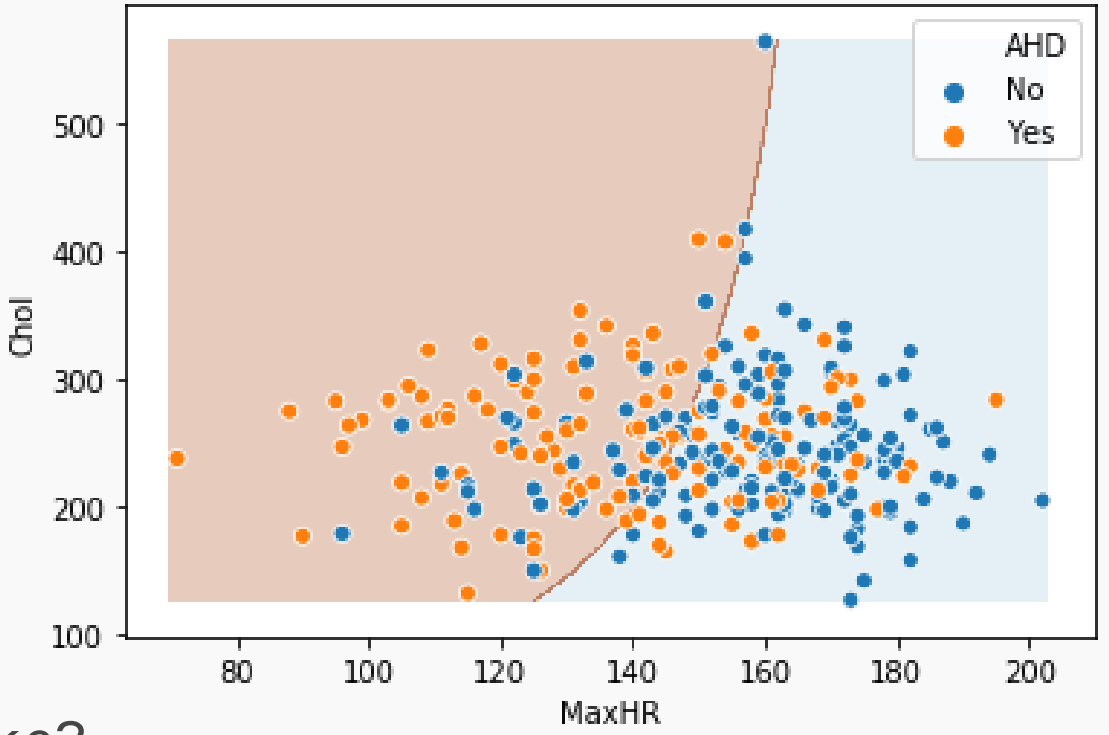
```
df_heart['Interaction'] = df_heart.MaxHR * df_heart.Chol

data_x = df_heart[['MaxHR','Chol', 'Interaction']]
data_y = df_heart['AHD']

logreg = LogisticRegression(penalty='none', fit_intercept=True)
logreg.fit(data_x, data_y);


print('Estimated beta1, beta2, beta3: \n', logreg.coef_)
print('Estimated beta0: \n', logreg.intercept_)
```

```
Estimated beta1, beta2, beta3:
 [[-0.00785835  0.02682656 -0.00015188]]
Estimated beta0:
 [5.70800455e-05]
```



What will the decision boundary look like?

In logistic regression, decision boundaries are not always structured to be linear!

2D Classification in Logistic Regression: Example #2

A logistic regression model was fit to predict Y = AHD from X_1 = MaxHR and X_2 = Chol, and X_3 = their interaction. Results are shown below:

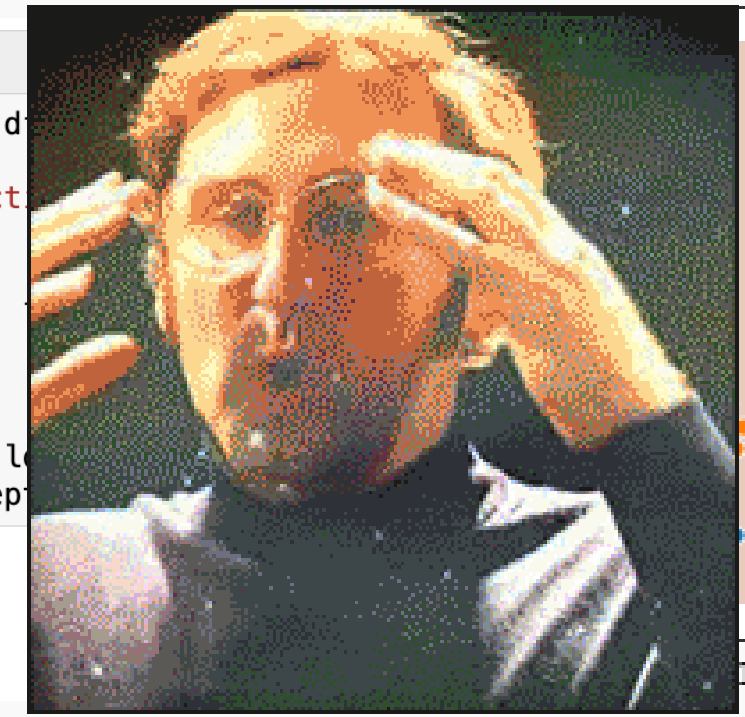
```
df_heart['Interaction'] = df_heart.MaxHR * df

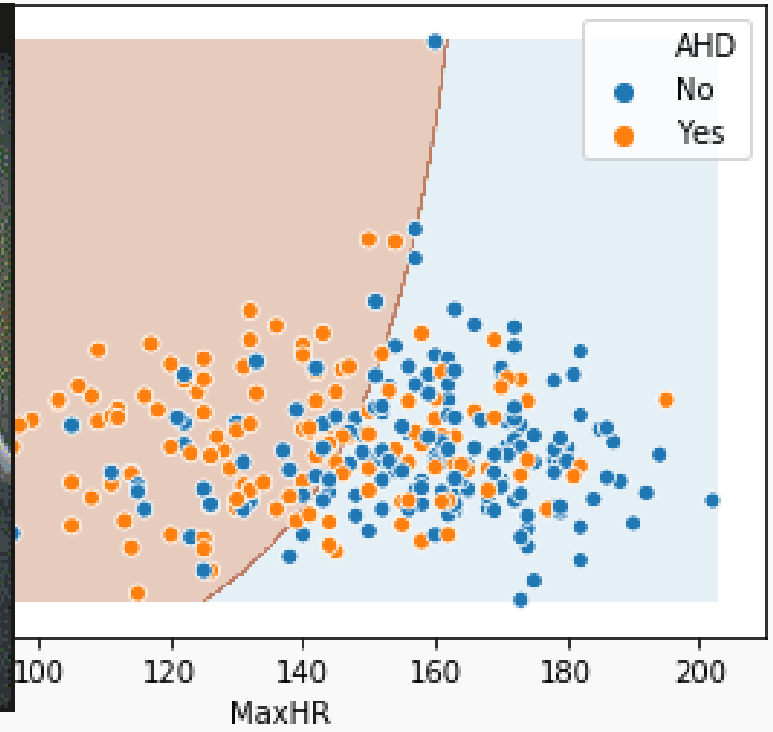
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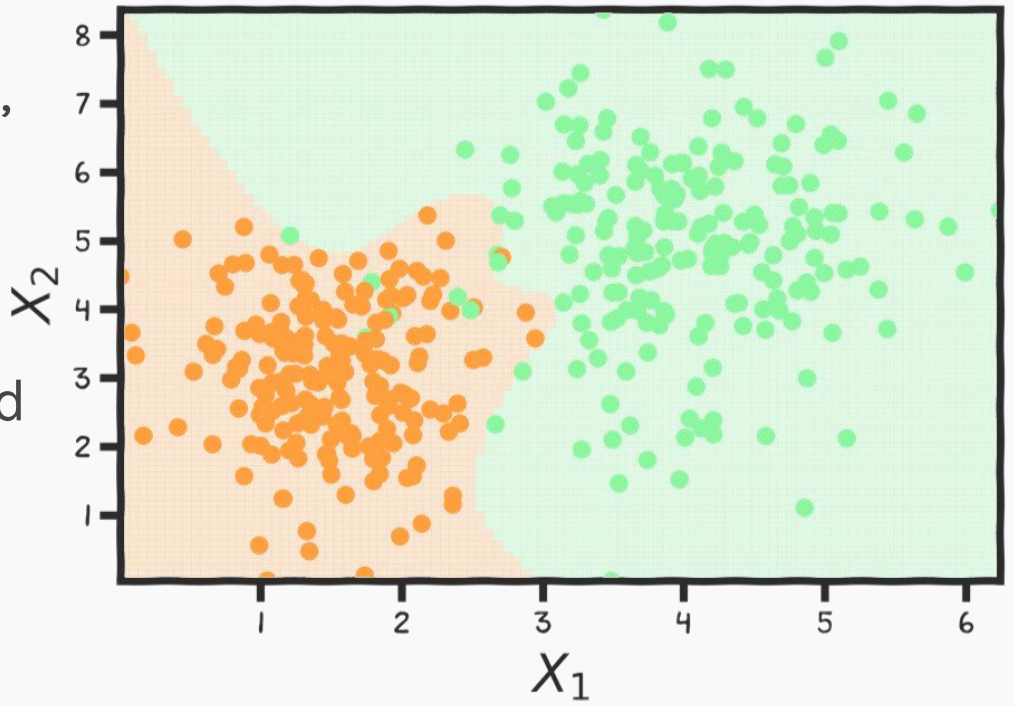


What will the decision boundary look like?

In logistic regression, decision boundaries are not always structured to be linear!

Polynomial Logistic Regression

We saw a 2-D plot last time which had two predictors, X_1, X_2. A similar one is shown here but the decision boundary is again not linear.



We can extend multiple Logistic Regression as we did with polynomial regression:

We transform the data by adding new predictors:

$$\tilde{x} = [1, \tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_M]$$

where $\tilde{x}_k = x^k$.

The polynomial Logistic Regression can be expressed as:

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \tilde{X}\beta$$

Geometry of Decision Boundaries (Logistic Regression)

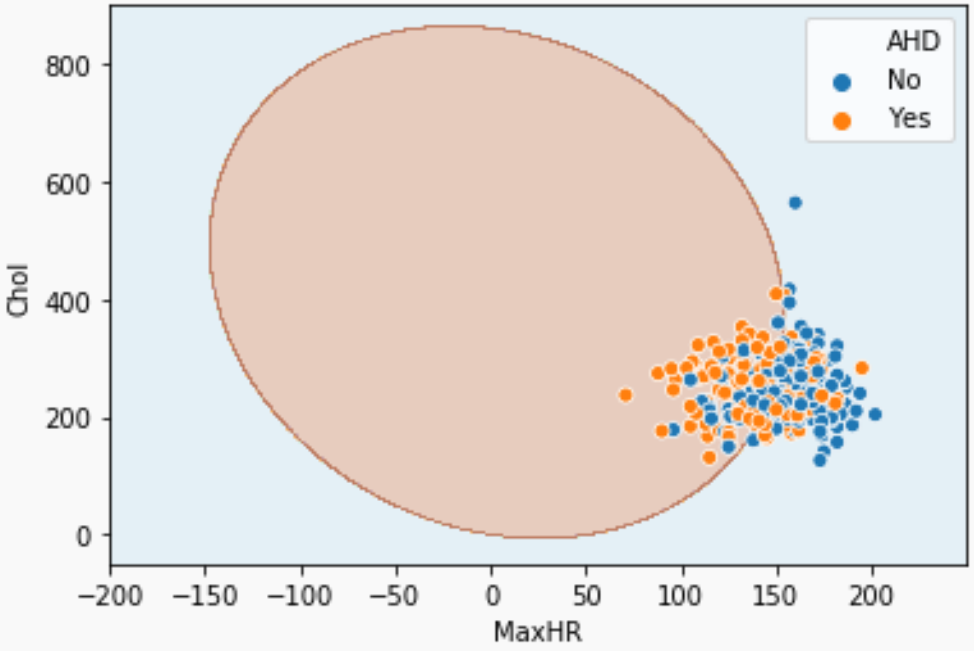
$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \tilde{X}\beta$$

Thus we can define our logistic regression model to achieve a desired geometry.

For example, what set for $X = f(X_1, X_2)$ should we choose if we want a *circular* decision boundary?

$$X = \{X_1, X_1^2, X_2, X_2^2, X_1X_2\}$$

What could be an alternative modeling approach (think outside of logistic regression)?



Decision Boundaries

